

Module 9: Ethics and Responsibility

Accountability, transparency, and team practice

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Primary sources: Osmani, Chapter 9; Thomas & Hunt, Pragmatic Teams.

Learning Outcomes

- Evaluate ethical risks in AI-assisted development.
 - Assess responsibility boundaries between engineers and AI systems.
 - Analyze bias and unintended consequences.
 - Justify responsible engineering practices.
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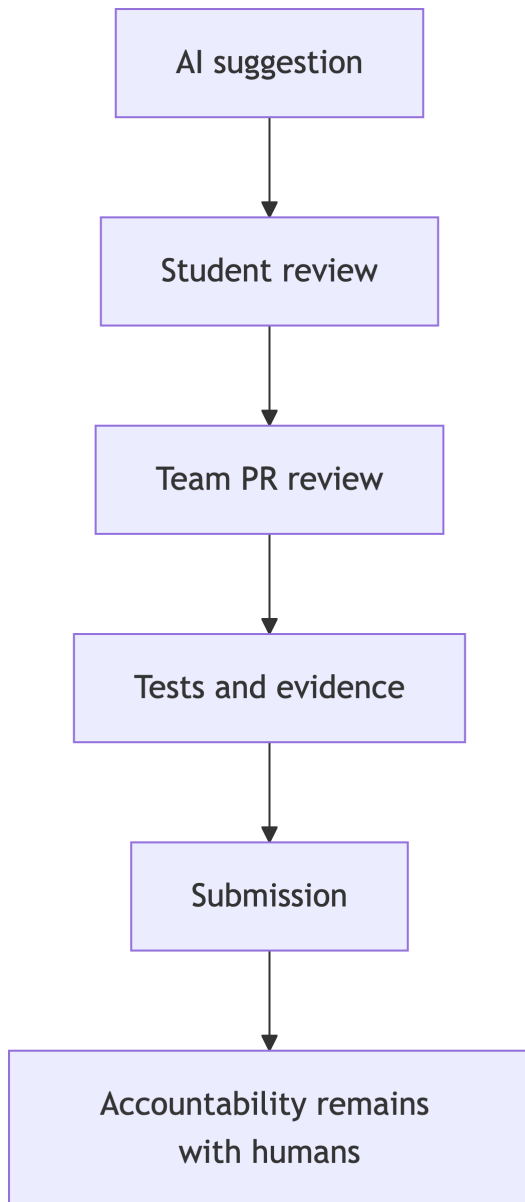
Responsibility Boundary

AI can assist, but it cannot be accountable.

The engineer or team remains responsible for:

- Submitted code.
 - Security impacts.
 - Documentation accuracy.
 - Disclosure of AI assistance.
 - User consequences.
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Accountability Flow



Transparency

Responsible AI use includes a clear artifact trail.

- What tool or mode was used?
 - What prompt or task was given?
 - What was accepted, modified, or rejected?
 - How was the result verified?
 - What uncertainty remains?
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IP and Provenance

AI-generated output can raise provenance questions.

Practical safeguards:

- Avoid asking for copyrighted code.
 - Be cautious with suspiciously specific output.
 - Prefer explanations, patterns, and tests over copied solutions.
 - Follow course and organizational policy.
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Bias and Consequences

Software can encode unfair assumptions.

AI can amplify them by producing common patterns without context.

Ask:

- Who could be harmed?
 - Whose data is missing?
 - What assumptions are hidden?
 - What feedback would reveal harm?
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Pragmatic Teams

Thomas and Hunt emphasize stable, communicative teams with shared standards.

For AI-assisted teams:

- Agree on AI usage rules.
 - Review AI-generated code carefully.
 - Share prompt patterns that work.
 - Do not let AI create invisible solo work.
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Controlled Collaboration

Team repositories should show:

- Issues or task board.
 - Feature branches.
 - Pull requests.
 - Review comments.
 - Test evidence.
 - AI usage notes.
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Lab 5 Development Connection

Coding agents increase ethical stakes because they act over multiple steps.

Controlled workflows need:

- Human checkpoints.
 - Reproducible runs.
 - Tool boundaries.
 - Reviewable artifacts.
 - Clear stop conditions.
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Practice Activity

Draft team AI rules:

- When is Copilot allowed?
 - What must be documented?
 - What requires peer review?
 - What data must never be pasted?
 - What counts as sufficient verification?
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Takeaways

- Responsible use is an engineering practice, not a slogan.
- Transparency supports grading, collaboration, and trust.
- Teams need explicit norms for AI-assisted work.